

Sabia Richards

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Sabia focuses on designing systems that can be verified to satisfy some property of a performance metric or logic of the result. She has worked with several math research groups, one of which worked on formalizing GPU cryptography algorithms in Rocq. More recently was designing logical guarantees into machine learning models and researching a framework for designing or proving properties about them. She believes proofs should be documented for verification and future modifications will likely require understanding their logic.

Professional Experience

Staff Machine Learning Engineer

AB InBev, July 2022 – September 2025

- Improved model training speed and memory usage in excess of 100x.
- Led teams building every part of the product's technology stack along with a close integration with users to determine what was useful and possible.
- Embedded mathematical guarantees of business properties in model architecture.
- Built a distributed framework to optimize marketing budget across tens of millions of dimensions with constraints and a low compute time requirement.
- Analyzed how floating point errors could break our guarantees, which errors we could fix, and which ones required a redesign.
- Helped design and participate in knowledge sharing from the senior developers.

Mobile CI/CD Engineer

ADT Security, August 2021 – April 2022

- Updated and maintained a Jenkins instance and associated infrastructure to move towards continuous development and testing for their mobile apps.
- Helped decide where automated gates could be used versus manual gating.
- Monitored telemetry from the deployed apps to find issues and summarize it and an approximate effort requirement before passing it along.

Software Developer

Zemoso, December 2018 – December 2020

- Led the refactoring to an MVVM Architecture and modern code design principle of an Android app.
- Developed a React Native cross-platform app and native bindings for native library access.
- Discovered optimizations and infrastructure improvements which consistently overperformed expectations.
- Managed, monitored, and created testing on the production deployments in the AWS kubernetes service.

Lead Software Developer

smartObjx, August 2017 – Dec 2018

- Designed a system for safe database migrations that had made testing difficult.
- Designed API Microservices and led a team of junior engineers in building them.
- Automated integration library generation so our partners could easily use any language in their projects. with little to no long term overhead.

Mobile SDK Developer

Apptimize, December 2015 – July 2017

- Built and deployed new features, third party integrations, and bug fixes for their mobile A/B and feature flag testing framework.
- Helped design and build the automated testing and deployment process which significantly improved developer velocity.

Speech Engineer

Bumpers, April 2015 – August 2015

- Researched using speech diarization in a cross-platform mobile app using the SHoUT algorithm and researched what other advancements were being made.
- Built a library for separating out blocks of speech without labeling speakers when full diarization proved not to be doable within the performance budget.

Mobile Developer

Fill My Blank, August 2014 – February 2015

- Developed and prepared for deploying a cross platform mobile app.
- Was successful at lowering network traffic and on-device resource usage.

Open Source Work

Magic the Gathering: Machine Learning (MtgML)

<https://github.com/CubeArtisan/mtgml>

- Created several models for predicting the next pick that a player would make in a given Magic the Gathering draft in an arbitrary environment.
- Built models for recommendations on additions and removals to cubes (collections of cards used for drafts).
- Tested approaches for model to build decks given the cards to choose from, had some promising results, but could use more research.

DraftbotOptimization

<https://github.com/ruler501/DraftbotOptimization>

- The first iteration of what would become the MtgML models using black box optimization techniques.
- Used heterogeneous compute libraries to allow accelerating optimization on whatever compute hardware was available.

CubeArtisan

<https://github.com/CubeArtisan/cubeartisan>

- A website made for managing Magic the Gathering cubes (collections of cards used for drafting).
- Provided a fully-fledged domain specific filtering language.
- Using the filtering language allowed for creating complicated assignments of cards to packs while maintaining randomness.
- Allowed calculating custom analytics on the complicated random distributions of cards into packs.
- Supported real time and asynchronous multiplayer drafting.

CubeArtisan Infrastructure

<https://github.com/CubeArtisan/cubeartisan-infrastructure>

- Deployment scripts and configurations for provisioning a kubernetes cluster for CubeArtisan.
- Supported multiple different environments for production and staging.
- Managed deployment of the MtgML models to be consumed by CubeArtisan and other sites.